Q7. For $n \geq 1$

(d). $$d_n = \frac{2n}{3n-1} - \frac{n^2+1}{2-n^2}$$

Use the algebra of limits theorem somehow.

Assuming the limits exist, the theorem will say:

$$\lim_{n\to\infty} d_n = \lim_{n\to\infty}\left(\frac{2n}{3n-1}\right) - \lim_{n\to\infty}\left(\frac{n^2+1}{2-n^2}\right)$$

Treating each one in turn.

$$\lim_{n\to\infty}\left(\frac{2n}{3n-1}\right)$$

$$= \lim_{n\to\infty}\left(\frac{2}{3-1/n}\right),$$ div. num. & denom. by n.

$$= \frac{2}{3-\lim_{n\to\infty}(1/n)},$$ using quotient and lin. combs. case of theorem.

$$= \frac{2}{3},$$ since $\lim_{n\to\infty} 1/n = 0$.

Similarly.

$$\lim_{n\to\infty}\left(\frac{n^2+1}{2-n^2}\right)$$

$$= \lim_{n\to\infty}\left(\frac{1+1/n^2}{2/n^2-1}\right),$$

$$= \frac{\lim(\quad)}{\lim(\quad)}, \text{ by } \cancel{\text{the}} \text{ quotient case of theorem}$$

$$= \frac{1+\lim 1/n^2}{2\lim(1/n^2)-1}, \text{ by lin. comb. case,}$$

$$= \frac{1}{-1}, \text{ as } \lim(1/n^2)=0$$

$$= -1$$

So $\lim_{n\to\infty} d_n = \frac{2}{3}+1 = 5/3$.

Q10. Consider the seq. defined by the recurrence relation.

$$x_0=2, \quad \forall n\geq 1 \quad x_n = \frac{1}{3-x_{n-1}}$$

Tips. ◦ Prove seq is bounded above by 2 and below by 0.

◦ Prove decreasing nature.

◦ Invoke M.C.T. to prove a limit exists.

◦ Use algebra of limits theorem to determine the limit.

Claim ~~[illegible]~~ $\forall n \geq 1 \quad x_n \leq 2$.

Proof By definition $x_0 = 2 \leq 2$ ✓

Assume $x_k \leq 2$.

then consider x_{k+1} ....

$$x_{k+1} = \frac{1}{3 - x_k}$$

Note that $3 - x_k \geq 1$, since $x_k \leq 2$.

So $x_{k+1} = \dfrac{1}{3 - x_k} \leq 1 < 2$.

Therefore, by induction, $\forall n \geq 1 \quad x_n \leq 2$.

Now we'll prove $\forall n \geq 1 \quad x_n > 0$.

Base case $x_0 = 2 > 0$ is true.

Assume $x_k > 0$, and we know $x_k \leq 2$.

then $3 - x_k \geq 1$.

$$\Longrightarrow x_{k+1} = \frac{1}{3 - x_k} > 0.$$

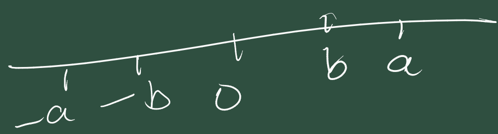
So by induction $\forall n \geq 0$ $x_n > 0$.

Claim. $\{x_n\}_{n=0}^{\infty}$ is always decreasing.

$x_0 = 2$, $x_1 = \frac{1}{3-2} = 1$

so clearly $\boxed{x_0 > x_1}$

Assume x_k x_{k-1} for some $k \geq 1$



$$\boxed{x_k < x_{k-1}}$$

$$\Longrightarrow -x_k > -x_{k-1}$$

$$\Longrightarrow 3 - x_k > 3 - x_{k-1} > 0$$

$$\Longrightarrow \frac{1}{3 - x_k} < \frac{1}{3 - x_{k-1}}$$

$\Rightarrow \boxed{x_{k+1} < x_k}$, by def. of the recurrence.

So by the principle of induction $\{x_n\}_{n=0}^{\infty}$ is decreasing.

By M.C.T. $\{x_n\}_{n=0}^{\infty}$ is convergent so

$$\lim_{n\to\infty} x_n = x, \text{ for some } x \in \mathbb{R}.$$

Now ~~by~~ use algebra of limits theorem after taking the limit of both sides of the recurrence relation.

$$x_{n+1} = \frac{1}{3-x_n}$$

$$\Rightarrow \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty}\left(\frac{1}{3-x_n}\right)$$

$$\Rightarrow \quad x = \frac{1}{3-x}, \text{ by algebra of limits theorem}$$

$$\Rightarrow \quad 3x - x^2 = 1$$

$$\Rightarrow \quad x^2 - 3x + 1 = 0.$$

$$\Rightarrow \quad x = \frac{3 \pm \sqrt{9-4}}{2}$$

$$= \frac{3 \pm \sqrt{5}}{2}.$$

So $x = \dfrac{3-\sqrt{5}}{2}$ or $\dfrac{3+\sqrt{5}}{2}$.

But $\dfrac{3+\sqrt{5}}{2} > 2$.

So $x \neq \dfrac{3+\sqrt{5}}{2}$.

Therefore $x = \frac{3-\sqrt{5}}{2}$.

and

$$\lim_{n\to\infty} x_n = \frac{3-\sqrt{5}}{2}.$$